

Erosive Burning of Aluminized Composite Propellants: X-Ray Absorption Measurement, Correlation, and Application

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The erosive burning effects in a small test motor loaded with highly aluminized practical composite propellants have been investigated in detail by using x-ray absorption diagnostics to measure the propellant local regression. The motor was specially designed to have two propellant slabs and was called a double-slab motor. Significant erosive burning was found in the motor for several combustion pressure levels and ranges of mass flux in the port. The x-ray diagnostic system enabled observation of the time-wise change in the distribution of burning propellant surface during motor operation. A modified Dickinson-type simple correlative equation, which includes the effects of mass flux, mass burning rate, combustion pressure, and motor scale, was derived from the results of DSM firing tests. Parameters of the correlative equation were finally determined by taking into account the experimental results of both cylindrical test motors and practical full-scale motors.

Nomenclature

A	=	constant in Eq. (3)
A_p	=	port area
B	=	constant in Eq. (3)
C	=	port perimeter
D_p	=	port hydraulic diameter
D_{p0}	=	initial value of D_p
D_{s0}	=	initial gap between two slabs
D_t	=	nozzle throat diameter
G	=	mass flux of port flow
G_{cr}	=	threshold value of G
$\bar{G}_{cr}$	=	dimensionless value of G_{cr}
G_{max}	=	port mass flux at downstream side
K_d, K_1, K_2	=	erosive burning coefficient in Eqs. (4), (6), and (7)
n	=	pressure exponent of burning rate
P	=	local pressure
P_{fwd}	=	pressure at upstream side
P_{ig}	=	pressure in igniter
r	=	linear burning rate
r_0	=	burning rate without erosive burning effect
t_p	=	time of pressure peak
V_s	=	output voltage of x-ray sensor
V_{s0}	=	value of V_s with no propellant

$\bar{w}$	=	average thickness of two slabs
w_1	=	thickness of top-side slab
w_2	=	thickness of bottom-side slab
x	=	axial distance from head end of slab
μ	=	x-ray absorption coefficient
ρ_s	=	density of propellant
σ_p	=	temperature sensitivity of r_0

Introduction

EROSIVE burning is a phenomenon in which the burning rate of solid propellant can be augmented by the interaction between reaction fields of the propellant and the gas flows parallel to the propellant burning surfaces. In the case of the solid propellant motors with internal burning grain, erosive burning appears when the mass flux in a grain port is beyond a threshold and causes chamber pressure to increase. Hence, motors with larger L/D (length to diameter ratio) enhance the chamber pressure rise due to erosive burning early in the firing.

Many theoretical and experimental studies have been conducted in erosive burning so far [1,2]. However, any correlative equations available for predicting the erosive effect with high accuracy have not been proposed due to the difficulties in developing a proper mathematical model and measuring the local burning rate in real motors. Some new techniques have recently been developed for measuring the burning rate by using ultrasonic waves [3,4] or microwaves [5,6]. On the other hand, the authors have developed a new x-ray diagnostic system, which can measure the local and instantaneous propellant thickness in a test double-slab motor (DSM) during the combustion period [7–10]. In this paper, the erosive burning effects of highly aluminized practical composite propellants have been measured by firing tests with the DSM. A new simple correlation based on Dickinson's [11] was derived by the curve fitting applied to both pressure and propellant surface regression histories. In addition, by taking into account the firing test results of other type motors, the best parameters in the correlative equation, which could be applied to typical practical full-scale motors, were obtained.

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Experimental Apparatus

A small test motor shown in Fig. 1 has two propellant slabs, therefore it is called the double-slab motor (DSM) [7–10]. The propellant slabs are placed face to face with each other. The DSM has a rectangular port area enclosed by the burning surfaces and sidewalls [fiber-reinforced plastic (FRP) liners], thus the two-dimensional flow field can be obtained. The range of mass flux is controlled by the initial gap between the two slabs. In this study, levels of the initial gaps between two slabs are set to 5 mm and 10 mm. The combustion pressure level was controlled by changing the nozzle throat diameter. A submerged nozzle is employed so as to control adhesion of combustion products to the inner surface of the nozzle throat [9].

Dimensions of a propellant slab are 250 mm in length, 50 mm in width, and 7.5 mm in thickness. The propellant supports are equipped with eight x-ray low-absorption aluminum windows, and they are arranged along the motor longitudinal axis with constant intervals (30 mm).

A schematic of the experimental setup is shown in Fig. 2. The experimental setup consists mainly of two x-ray generators and the DSM. The x-ray generators are placed in tandem above the DSM, and the distance between the x-ray generators and the center axis of the DSM is 700 mm. X rays are supplied from generators and detected by eight sensors (scintillation counters) after transmitting the two propellant slabs and combustion gases between two burning surfaces of slabs. The instantaneous propellant thickness is obtained by referring to an “x-ray intensity vs propellant thickness” calibration, in-place experimentally determined.

The x-ray intensity detected at each measurement position is represented by each sensor output voltage, which is amplified with the DC amplifier. The correlation between the x-ray intensity detected and the total thickness of the two propellant slabs obeys Lambert's law.

$$V_s = V_{s0} \exp[-\mu(w_1 + w_2)] \quad (1)$$

where V_s , V_{s0} , and μ represent the sensor output voltage proportional to the x-ray intensity, sensor output voltage with no propellant, and absorption coefficient, respectively. The coefficient μ is not constant for white x rays, but decreases as the x-ray transmitted distance increases. The local thickness of one slab $\bar{w}$ is defined as the average of the thicknesses of the two slabs.

$$\bar{w} = \frac{1}{2}(w_1 + w_2) \quad (2)$$

According to the data of in situ calibration, μ was almost a linear function of $2\bar{w}$.

$$\mu = A + B(2\bar{w}); \quad A, B = \text{const.} \quad (3)$$

By statistical analysis of calibration data [9], the measurements of averaged thickness $\bar{w}$ had an accuracy of ± 0.1 mm. The results of

calibration and correlation between V_s and $2\bar{w}$ are shown in Fig. 3. It has been verified that the propellant thickness distributions in the lateral direction and local thickness of two slabs along the motor central axis were almost uniform by the interrupted combustion test [9]. The result of this verification test shows that the change in the x-ray intensity due to presence of the combustion gas including condensed particles is negligible [8].

Two types of aluminized composite propellants, which are used in practical motors, were tested in this experiment. Main components of both propellants are HTPB, AP, and Al. One of them includes a small amount of additive Fe_2O_3 . Therefore the levels of linear burning rate are different; 4.9 mm/s (@4.9 MPa) for lower burning rate propellant (propellant no. 1) and 9.2 mm/s (@4.9 MPa) for higher burning rate propellant (propellant no. 2). Range of combustion pressure level chosen was the same as that used with practical rocket motors loaded with each propellant; 2–8 MPa for lower burning rate propellant and 4–10 MPa for higher burning rate propellant. The properties of propellants and experimental conditions are shown in Tables 1 and 2.

Experimental Results of DSM

A typical pressure-time curve and history of averaged surface regression of the slabs measured by a DSM firing test (propellant no. 1, $P = 5$ MPa; $D_{s0} = 5$ mm) are shown in Fig. 4. The igniter was so designed that the DSM was ignited just at the igniter burnout [9]. As seen in this figure, igniter pressure P_{ig} fell when the upstream-side pressure of DSM P_{fwd} started to rise. Thus, the combustion gas from the igniter was not present in the motor during the steady combustion period. Erosive burning effect is usually significant in the early phase

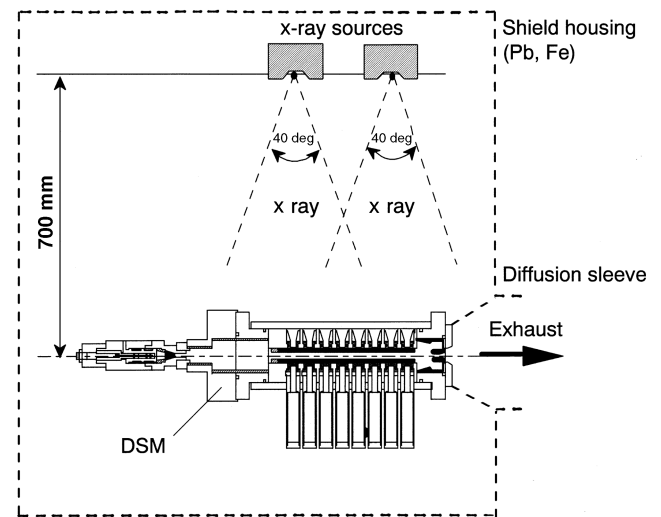


Fig. 2 Schematic of experimental setup.

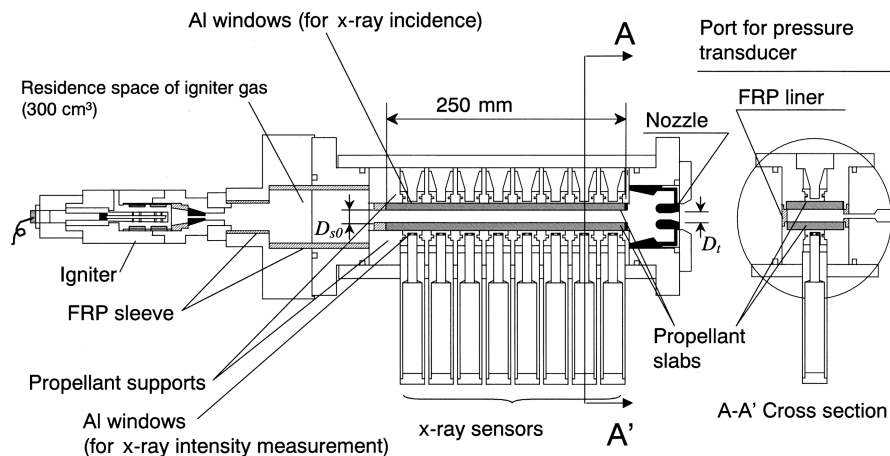


Fig. 1 Double-slab motor (DSM).

of motor operation with a smaller port area, so that a pressure peak appears in the transition period from the ignition transient to steady burning. When the pressure peak appears at the time t_p after motor ignition, recession of combustion surfaces averages 1.3 mm from the initial surface and 18% of the total loading mass has been burned. In the steady burning period just after the t_p , it is clearly seen that the combustion surface is inclined to the longitudinal axis x . This indicates the erosive burning effect because burning rate becomes higher toward the aft end of the grain. Although a submerged nozzle was employed, melted aluminum or alumina adhered to the inner surface of the nozzle throat. Therefore pressure history without the erosive burning effect may be slightly progressive contrary to the present case.

Figure 5 shows the influence of the initial gap between two propellant slabs on the erosive burning appearance. Two levels of the initial gap ($D_{s0} = 5$ mm and 10 mm) were tested. In the case of a small initial gap, the pressure curve is regressive due to the erosive burning effect. The contour of burning surface regression shows that the burning rate in the downstream side is higher than in the upstream side. Since the erosive burning effect appears significantly in the ignition transient region, the slope of burning surface is inclined along the motor axis. On the other hand, for $D_{s0} = 10$ mm, the pressure curve is neutral and the counter of burning surface is almost flat. This shows less erosive burning effect due to the decrease in port mass flux and scale effect.

The difference in the erosive burning between two propellants is shown in Fig. 6. It is seen from this figure that the magnitude of burning rate increment of propellant no. 2 is much smaller than that of propellant no. 1. This reflects the well-known fact that the erosive

burning effect decreases with the increase in normal burning rate. Propellant composition may also affect the erosive burning characteristics through the difference in flame structure even if normal burning rate is the same.

Evaluation of Erosive Burning Effect

For analyzing the erosive burning characteristics, the pressure curves and burning surface regression contours obtained in a series of DSM experiments are compared with calculations. For this purpose, a suitable correlative equation of erosive burning is needed. Many correlative equations have been proposed so far and are reviewed in [1,2]. Recently, Mukunda and Paul [12] proposed a new equation that was derived from empirical knowledge and theories of fluid dynamics. The authors have also studied theoretically erosive burning to develop a correlative equation for practical use [13,14]. However, any satisfactory correlative equation, which is appropriate enough from the view points of both theory and experiment, has not been obtained due to the complexities of port flow and combustion mechanisms in real motors. Therefore, Dickinson's correlation is employed in this study due to its simplicity and practicality. The original Dickinson's correlation is expressed as follows [11].

$$\frac{r}{r_0} = 1 + K_d G \left(1 - \frac{G_{cr}}{G} \right) \quad (4)$$

where subscript 0 represents no erosive burning condition and G_{cr} is the threshold condition.

This correlation emphasizes that the erosive burning effect is mainly controlled by the mass flux in the port and when its value exceeds a threshold value, the erosive burning effect becomes significant. This seems to be oversimplified from the viewpoint that the erosive burning occurs by the interaction between port flow and solid propellant burning.

According to the preliminary study by the authors, the value of G_{cr} is almost proportional to mass burning rate independent of propellant composition and combustion pressure [14,15].

$$G_{cr} = 10\rho_s r_0 \quad (5)$$

First, taking this result into account, nondimensional mass flux $\hat{G} = G/\rho_s r$ is introduced, and mass flux G is replaced by $\hat{G}$ only in the brackets of Eq. (4). This means that erosive burning occurs when $\hat{G}$ exceeds a threshold value $\hat{G}_{cr}$, instead of G in the original Dickinson's correlation. Although the physical basis for this procedure is not rigid, the experimental result by us mentioned

Table 1 Propellant properties

	Propellant no. 1	Propellant no. 2
Composition, % by mass		
Oxidizer (AP)	69	68
Binder (HTPB)	17	12
Metal fuel (Al)	14	20
Additive (Fe ₂ O ₃)	—	~0.5
Density ρ_s , g/cm ³	1.7	1.8
Linear burning rate r_0^a , mm/s	4.9	9.2
Pressure exponent n	0.3	0.4
Temp. sensitivity σ_p , %/K	0.155	0.076
Adiabatic flame temp. ^b , K	3041	3530
Mean molecular mass ^b , g/mol	25.4	29.6
Specific heat ratio ^b	1.19	1.17

^a@4.9 MPa, 20 deg.

^bFrozen equilibrium calculation @5 MPa.

Table 2 Experimental conditions

Pressure level ^a P , MPa	Propellant no. 1			Propellant no. 2		
	Initial gap between slabs D_{s0} , mm	Nozzle throat diameter D_t , mm	Maximum port mass flux ^a G_{max} , kg/m ² s	Initial gap between slabs D_{s0} , mm	Nozzle throat diameter D_t , mm	Maximum port mass flux ^a G_{max} , kg/m ² s
2	5	12.2	667	—	—	—
2	10	12.2	334	—	—	—
2.5	5	11.6	709	—	—	—
2.5	—	—	—	—	—	—
3	5	10.7	745	—	—	—
3	—	—	—	—	—	—
4	5	9.8	805	5	13.5	1542
4	10	9.8	402	10	13.5	771
5	5	9.2	855	—	—	—
5	10	9.2	427	—	—	—
6	5	8.7	898	5	12.2	1814
6	10	8.7	449	10	12.2	907
8	5	7.9	970	5	11.2	2035
8	—	—	—	10	11.2	1017
10	—	—	—	5	10.5	2225
10	—	—	—	10	10.5	1112

^aNo erosive burning.

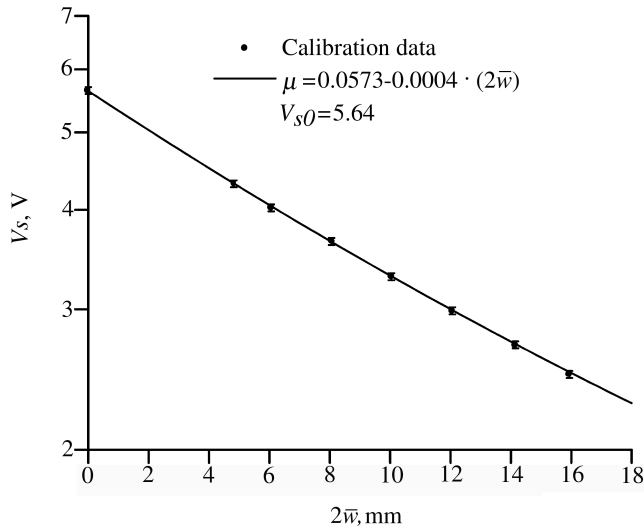


Fig. 3 Correlation between V_s and slab thickness $2\bar{w}$.

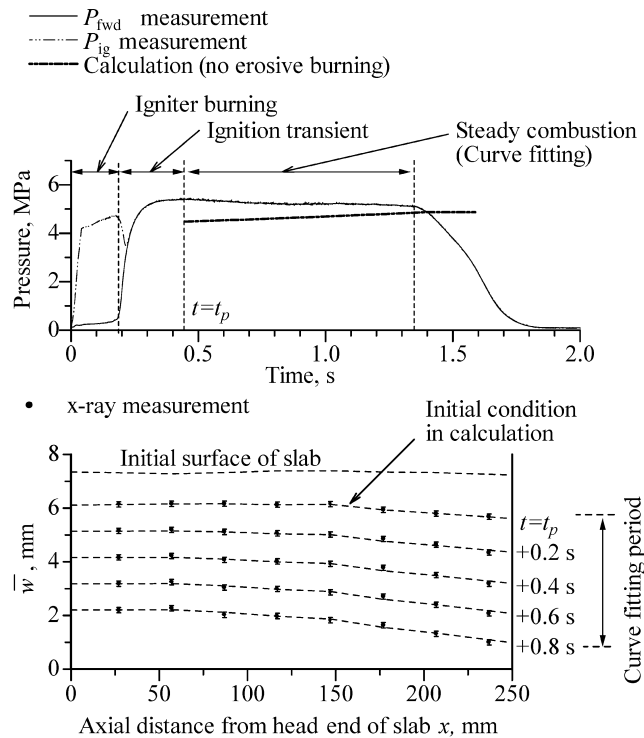


Fig. 4 Typical pressure-time curves (upper) and local surface regression histories (lower).

previously and also the results by other researchers seem to support this at least qualitatively.

Second, another important modification is introduced here. Although the original Dickinson's correlation assumes that the erosive burning increment is proportional to G , this assumption is relaxed and replaced by G^γ instead of G .

Many theoretical or experimental studies indicate that the erosive burning ratio depends on several factors other than propellant composition, that is, port diameter, curvature of surface, roughness of surface, pressure gradient, and local pressure [1]. Therefore, the erosive burning coefficient K_d in Eq. (4) may not be constant, but changes due to these effects. Among these factors, scale effect is one of the important factors and included in many practical correlations. This effect is usually represented by the port diameter. When a motor has a noncircular port, hydraulic diameter defined by $D_p (=4A_p/C)$ can be used as an effective port diameter. Although scale effect is often expressed by a powerlike function of the port diameter,

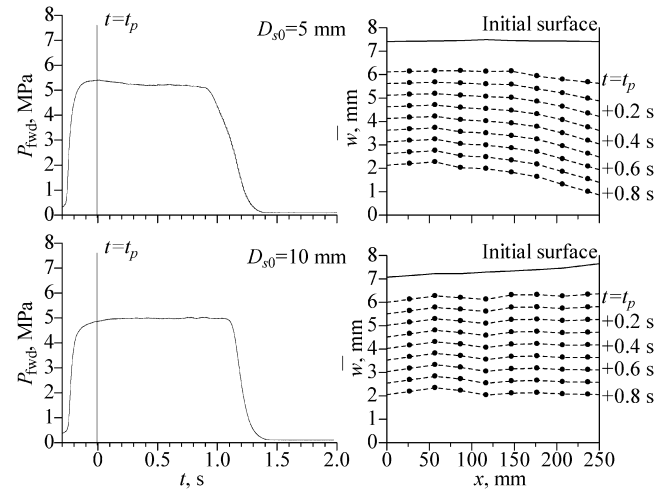


Fig. 5 Measured pressure-time curve (left) and surface regression history (right) of propellant no. 1 at 5 MPa.

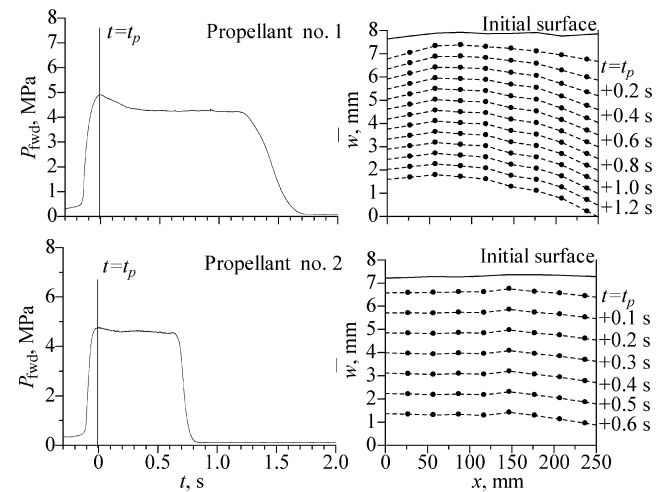


Fig. 6 Comparison of the erosive burning effects between two propellants at 4 MPa.

different values were suggested for the exponent of D_p so far [1,14]. In this study, this situation is retained, and the exponent was set to -0.2 in view of turbulent theory. From the preceding considerations, the Dickinson's correlation is modified into the following dimensional form.

$$\frac{r}{r_0} = \begin{cases} 1 + K_1 D_p^{-0.2} \left[G \left(1 - \frac{\hat{G}_{cr}}{\hat{G}} \right) \right]^\gamma & ; \hat{G} > \hat{G}_{cr} \\ 1 & ; \hat{G} \leq \hat{G}_{cr} \end{cases} \quad \hat{G}_{cr} = 10 \quad (6)$$

Histories of pressure and surface regression are calculated by solving one-dimensional, steady internal flow equations with mass and energy addition from the burning surface and variable port area [10]. Here, mass and energy addition to the flow are calculated using Eq. (6). The remaining parameters in Eq. (6) are so determined that the pressure and local surface regression were consistent with each measurement.

DSM Results

The best selection of the two parameters γ and K_1 in Eq. (6) is rather cumbersome. Therefore, as for γ , $\gamma = 1$ was employed first for DSM cases. As will be explained later, this selection is adequate for a motor with a low or moderate mass flux such as DSM, but is insufficient for a motor with a much higher mass flux.

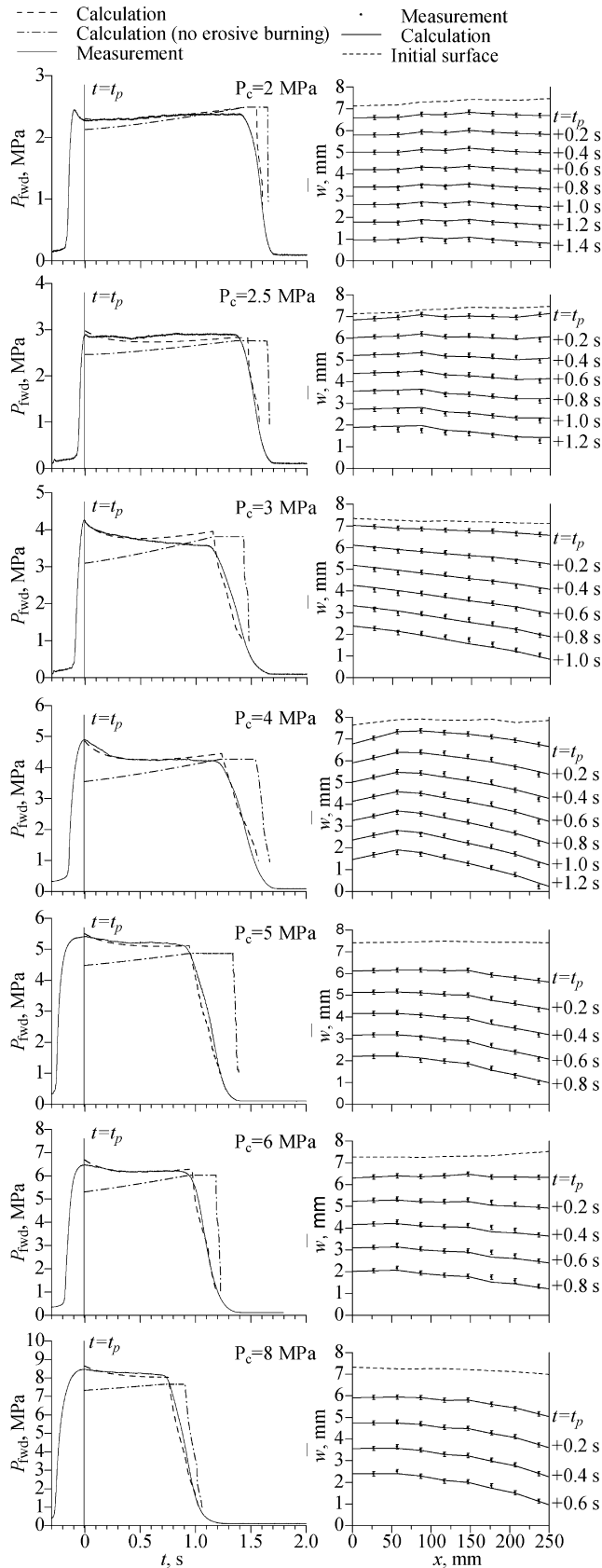


Fig. 7 Measured and calculated data on both pressure-time curve and surface regression history for propellant no. 1 and $D_{s0} = 5$ mm.

The results of numerical calculation by using the best value of K_1 are shown in Figs. 7–10 with the experimental results. Calculation was started from t_p with the initial condition of port area distribution given by x-ray measurement. As already stated, nozzle throat

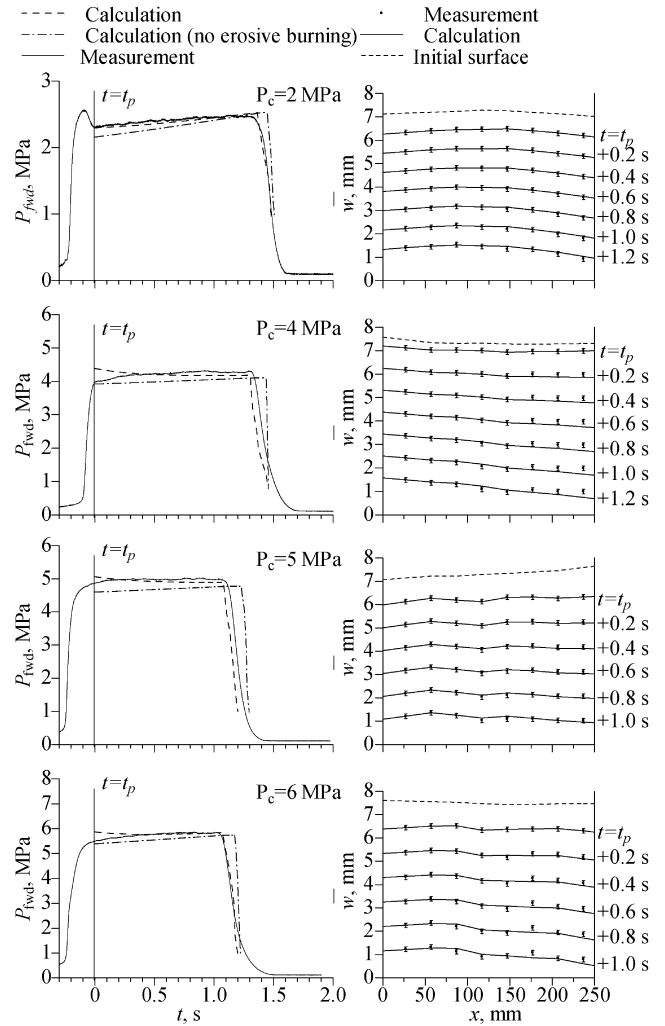


Fig. 8 Measured and calculated data on both pressure-time curve and surface regression history for propellant no. 1 and $D_{s0} = 10$ mm.

diameter is changed during the steady combustion period due to the adhesion of the condensed Al/Al₂O₃ mixture; this effect should be treated properly for the numerical simulation. In the practical calculation, the nozzle throat diameter was assumed proportional to time based on the experiments [15].

In Fig. 11, the best values of K_1 obtained for each experiment are plotted against combustion pressure. Error bars about pressure axis denote the range of pressure change during the steady combustion period. This figure shows that the level of K_1 differs to a great extent depending on the propellant type. In addition, it is also clear that K_1 has pressure dependence between 2 MPa and 10 MPa for both propellants. In the case of propellant no. 1, K_1 decreases drastically as combustion pressure decreases below 4 MPa. On the other hand, for propellant no. 2, it is clearly seen that K_1 decreases as combustion pressure increases above 8 MPa. In the intermediate region between 4 MPa and 8 MPa, in which steady-state combustion pressure generally falls, pressure dependence of K_1 could be approximated by the following power function of pressure with a negative exponent.

$$K_1 = K_2 P^{-0.7}$$

$$K_2 = \begin{cases} 1.02 \times 10^{-3} & \text{; Propellant No. 1} \\ 2.47 \times 10^{-4} & \text{; Propellant No. 2} \end{cases}, \quad [\text{m}^{2.2} \text{ s MPa}^{0.7} / \text{kg}] \quad (7)$$

$$\gamma = 1.0$$

This pressure dependence is qualitatively similar to Kreidler [16].

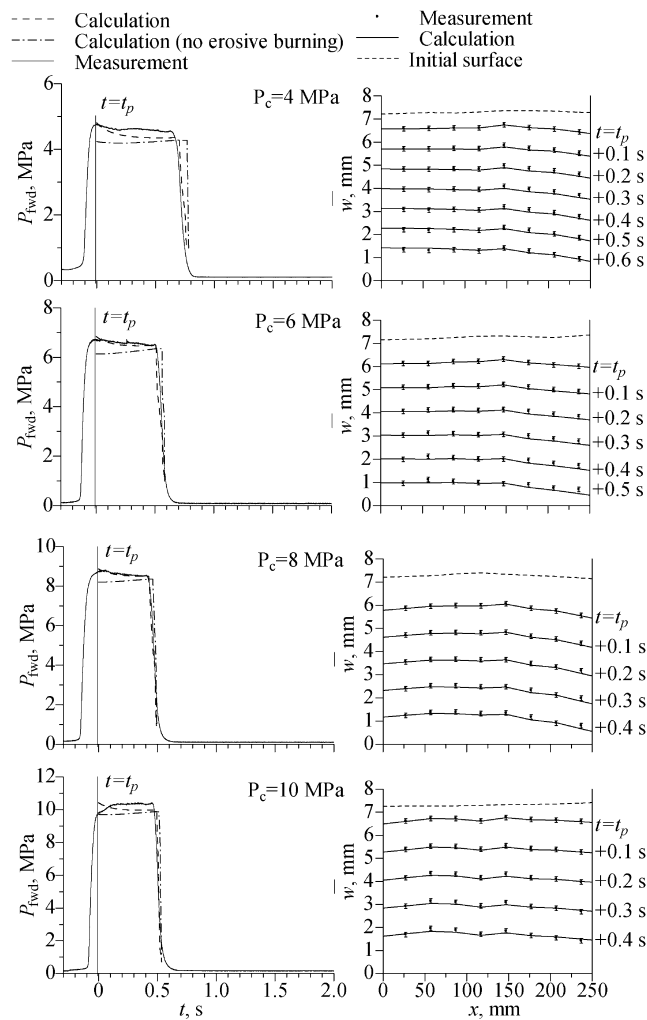


Fig. 9 Measured and calculated data on both pressure-time curve and surface regression history for propellant no. 2 and $D_{s0} = 5$ mm.

Results of Practical Motors

To investigate whether the proposed equation can predict the erosive burning effect properly in practical motors, firing tests of high L/D cylindrical motors loaded with propellant no. 1 (lower burning rate propellant) were also carried out. Three types of motor with different dimensions were used for this purpose. The properties of the motors are shown in Table 3. Type A has high mass flux (~ 2000 kg/m² s), which falls in the same range as a typical sounding rocket motor. The initial port diameter of this test motor is 40 mm, and grain length is 1600 mm. The ranges of mass flux of the other two motors (types B and C) are the same order as the DSM (~ 700 kg/m² s). The initial port diameter of type B is 40 mm ϕ , and that of type C is 60 mm ϕ . Nozzle throat diameter was so adjusted that the initial pressure level without the erosive burning effect was the same. Pressure histories were calculated by the same code applied to the DSM, but burned thicknesses in the ignition transient period were neglected because of lack of reliable data. Pressure histories calculated with Eq. (7) at the upstream side of the motor are compared with the measurements in Fig. 12. Because of minimized igniter gas flow rate, ignition of main motors occurs after igniter burnout. So, the ignition peaks that appeared are not due to the igniter gas addition. For these peaks, unsteady burning of propellant might play an important role in addition to erosive burning. Therefore, these peaks were not considered when evaluating the erosive burning effect.

For the cases of types B and C, calculations agree well with the measurements. These results support the adequacy of pressure and hydraulic diameter dependences in Eqs. (6) and (7). However, for the case of type A, the calculated result deviates from the measurement,

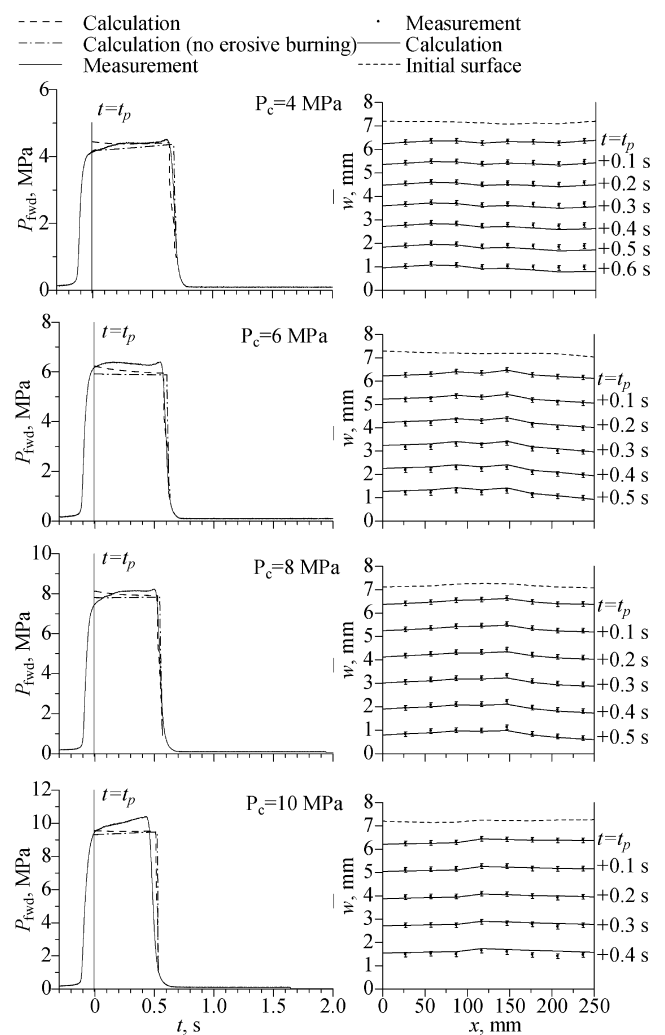


Fig. 10 Measured and calculated data on both pressure-time curve and surface regression history for propellant no. 2 and $D_{s0} = 10$ mm.

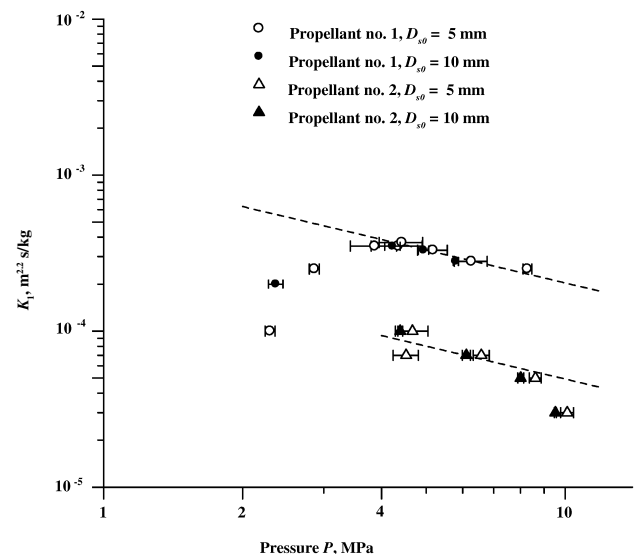


Fig. 11 Relations between K_1 and pressure P .

and the predicted pressure level is lower than that of the measurement. All of these results indicate that the correlative equation obtained by DSM firing tests can be applied to cylindrical motors that have the same range of port mass flux as the DSM, but the measured erosive burning effect in the high mass flux region

- Calculation with Eqs. (6,7)
 -.-.-.- Calculation (no erosive burning)
 ——— Measurement P_{fwd}
 -.-.-.- Measurement of igniter pressure P_{ig}

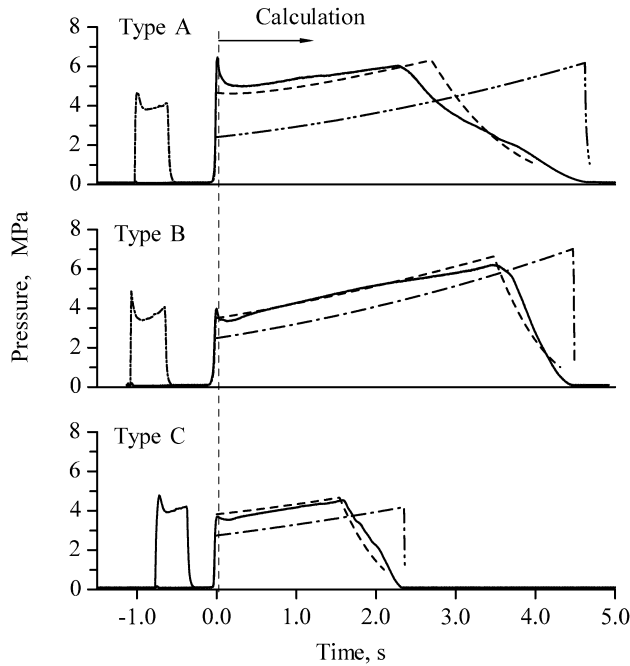


Fig. 12 Measured and calculated pressure histories of cylindrical motors.

- Calculation with Eqs. (6,8)
 -.-.-.- Calculation (no erosive burning)
 ——— Measurement P_{fwd}
 -.-.-.- Measurement of igniter pressure P_{ig}

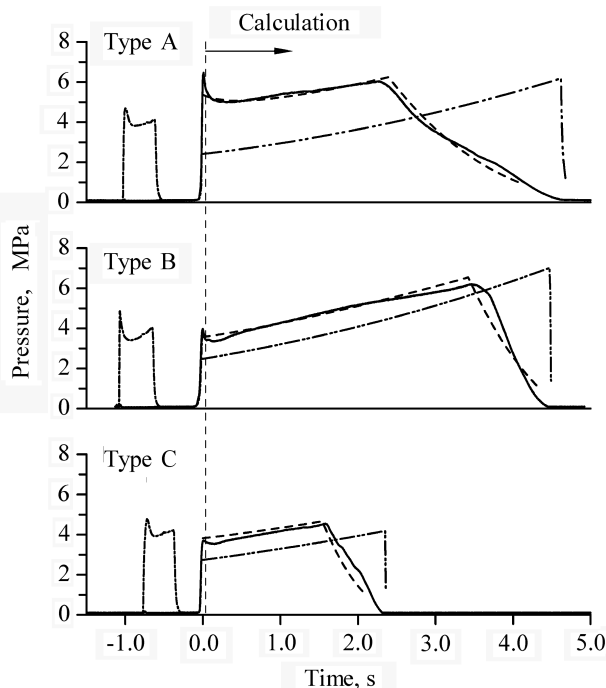


Fig. 13 Measured and calculated pressure histories of cylindrical motors by using Eqs. (6) and (8).

Table 3 Properties of cylindrical motors

	Type A	Type B	Type C
Nozzle throat diameter D_t , mm	34	23	34
Initial port diameter D_{p0} , mm	40	40	60
Outer diameter of grain, mm	80	80	80
Length of grain, mm	1680	840	1260
Maximum mass flux in port ^a G_{max} , $\text{kg/m}^2 \text{ s}$	2000	700	730

^aCalculated including the erosive burning effect.

($G > 700 \text{ kg/m}^2 \text{ s}$) is stronger than the prediction. This suggests that linear dependency of the erosive burning effect on the mass flux G is not satisfactory. So, the value of the exponent γ in Eq. (6) is reselected so as to satisfy the erosive burning effect in the high mass flux region without deteriorating in the moderate mass flux region, such as with the DSM. As the result, $\gamma = 1.2$ was found to be the best for both propellants. The best value of the coefficient K_2 was simultaneously determined as

$$\gamma = 1.2,$$

$$K_2 = \begin{cases} 3.4 \times 10^{-4} & ; \text{propellant No. 1} \\ 7.0 \times 10^{-5} & ; \text{propellant No. 2} \end{cases}, [\text{m}^{2.6} \text{ s}^{1.2} \text{ MPa}^{0.7} / \text{kg}^{1.2}] \quad (8)$$

In the low mass flux region ($G < 700 \text{ kg/m}^2 \text{ s}$), the value of $G^{1.2}$ is nearly equal to $G^{1.0}$ by multiplying a suitable constant. So, the new set of parameters given by Eq. (8) could be applied to the DSM and types B and C motors with similar accuracy to Eq. (7). The pressure histories calculated again with Eq. (8) for each cylindrical motor are shown in Fig. 13. The calculations for the three motor types, including type A, agree well with the measurements.

Furthermore, the obtained correlation is also applied to practical motors (S-310 and M-14). The S-310 booster [of the Japan Aerospace Exploration Agency (JAXA)] shown in Fig. 14 is loaded with propellant no. 1, and the range of port mass flux is the same order as the type A motor described previously. The grain is designed uniquely. The shape of the port area has been changed in the middle of the total grain length. So, the pressure level changes steeply during the steady burning period. Outer dimensions of the S-310 are 310 mm in diameter and 5157 mm in length, including nozzle assembly. The M-14 motor (JAXA) shown in Fig. 15 is used as the first stage (B1) motor of the M-V launch vehicle. This motor is loaded with propellant no. 2, and consists of two segments. Each segment has seven deep slits cutting into the outer surface from the inner surface of the central circular port area, but geometrical details of the port area of each segment are different. The range of the port mass flux of the M-14 is about $2000 \text{ kg/m}^2 \text{ s}$, and the averaged steady-state pressure is 4.8 MPa. Outer dimensions of the M-14 are

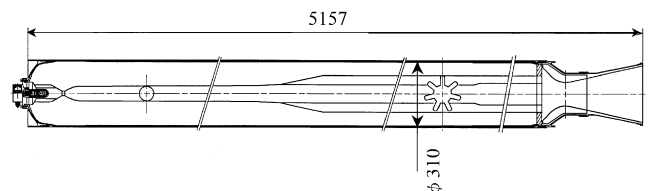


Fig. 14 Schematic of the S-310 booster (JAXA).

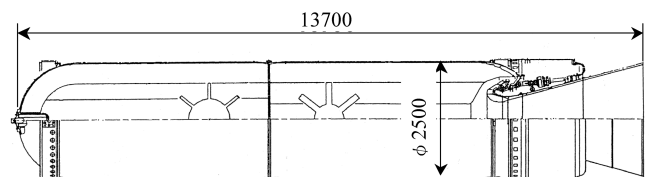


Fig. 15 Schematic of the M-14 motor (JAXA).

2500 mm in diameter and 13,700 mm in length, including nozzle assembly.

Calculation of the pressure history has been done using the one-dimensional steady internal flow solver, which is basically the same as that mentioned previously. In this calculation, Eqs. (6) and (8) were applied to both motors. The calculated pressure histories of each motor are shown in Figs. 16 and 17. As seen in these figures, discrepancies between prediction and experiment are small if they are compared with the pressure increment due to the erosive burning

effect. Nevertheless, the remaining differences cannot be neglected. To get better agreement, the value of K_2 was readjusted. When the best values [$K_2 = 2.8 \times 10^{-4} \text{ m}^{2.6} \text{ s}^{1.2} \text{ MPa}^{0.7} / \text{kg}^{1.2}$ (for S-310) and $K_2 = 4.5 \times 10^{-5} \text{ m}^{2.6} \text{ s}^{1.2} \text{ MPa}^{0.7} / \text{kg}^{1.2}$ (for M-14)] were used, the calculation agreed well with the measurements. This result could be explained as follows. It is reported that the erosive burning effect is not uniform in the cross section of star-shaped grain [11,17]. The difference in the value of K_2 between the full-scale motors and cylindrical motors or DSM might be due to the estimating error of normal burning rate or nonuniformity of the erosive burning effect described previously.

Discussions

Mechanisms in the chemical reaction zone (combustion zone) of a composite propellant are influenced by the dominant turbulent flow above the propellant surface. Generally speaking, heat flux to the burning surface increases because diffusion flames in the gas phase approach the solid surface by turbulent mixing, hence the burning rate increases with heat flux. This is the phenomenological understanding of erosive burning. Although the distance between the flame and the solid surface (flame height) decreases with pressure increasing, characteristics of turbulent flow field would be basically independent on pressure. As a result, the erosive burning effects become relatively weak as combustion pressure increases. The flame height of the higher burning rate and higher adiabatic temperature propellant (no. 2) would be less than that of the lower burning rate propellant (no. 1). So, the erosive burning effect of the higher burning rate propellant (no. 2) should be weak compared with the lower burning rate propellant (no. 1). The values of the coefficients and pressure dependence given by Eqs. (7) and (8) are consistent qualitatively with the characteristics described previously.

Temperature sensitivity σ_p is defined as the percent change of burning rate per degree change in propellant temperature. Interrelations between σ_p , diameter of AP particles, and activation energy of gas phase reaction have been described by Cohen and Flanigan [18]. They also showed that the port flow above combustion surface changes σ_p . For this reason, the distance between the solid surface and the diffusion flame could be characterized by σ_p . Therefore, σ_p could be adopted as one of the parameters, which characterizes the erosive burning effect. The values of σ_p of each propellant are shown in Table 1. It is seen that the value of propellant no. 1 is about twice that of no. 2. This suggests that the value of K , which represents the magnitude of the erosive burning effect, increases with temperature sensitivity σ_p .

As shown in Fig. 11, in the case of lower burning rate propellant (no. 1), the erosive burning effect decreases drastically as combustion pressure decreases in the pressure range less than 4 MPa. This phenomenon cannot be explained by the interrelation between flame height and pressure. In this region, the maximum values of port mass flux including erosive burning effect at t_p were larger than $500 \text{ kg/m}^2 \text{ s}$ and sufficient to cause erosive burning. Therefore, this effect cannot be explained by a lack of mass flux range. As far as the authors know, there is no data or theory that can explain this phenomenon.

Although the erosive burning ratio r/r_0 has been described as proportional to G or $G^{0.8}$ in the past representative research [11,19], it has been found here that the erosive burning ratio might be proportional to $G^{1.2}$.

Conclusions

Erosive burning effects of two typical composite propellants were observed and measured directly by an x-ray diagnostic system in a specially designed double-slab motor (DSM). The x-ray absorption diagnostic was shown to be effective for measuring a burning solid propellant thickness, and enabled observation of the time-wise change in the distribution of burning propellant surface during a motor operation. A modified Dickinson-type simple correlative equation was proposed based on the experimental results of the

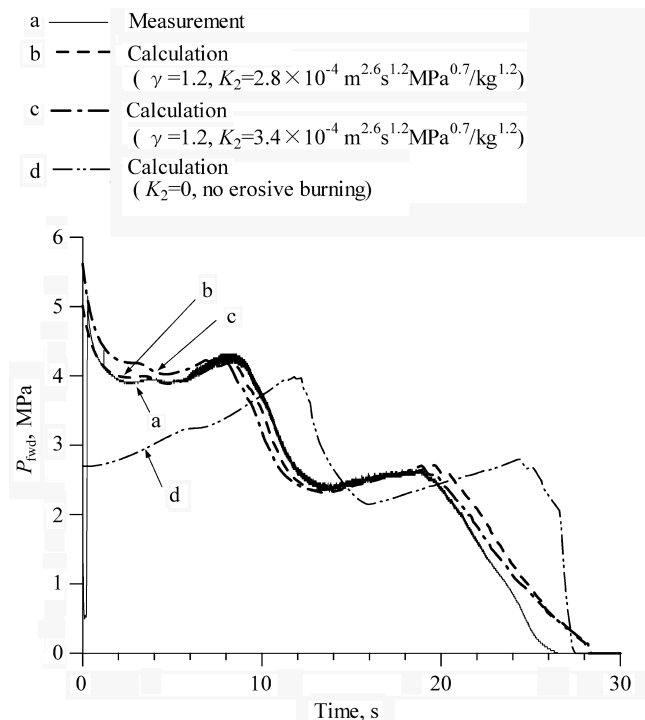


Fig. 16 Measured and calculated pressure histories of the S-310 booster.

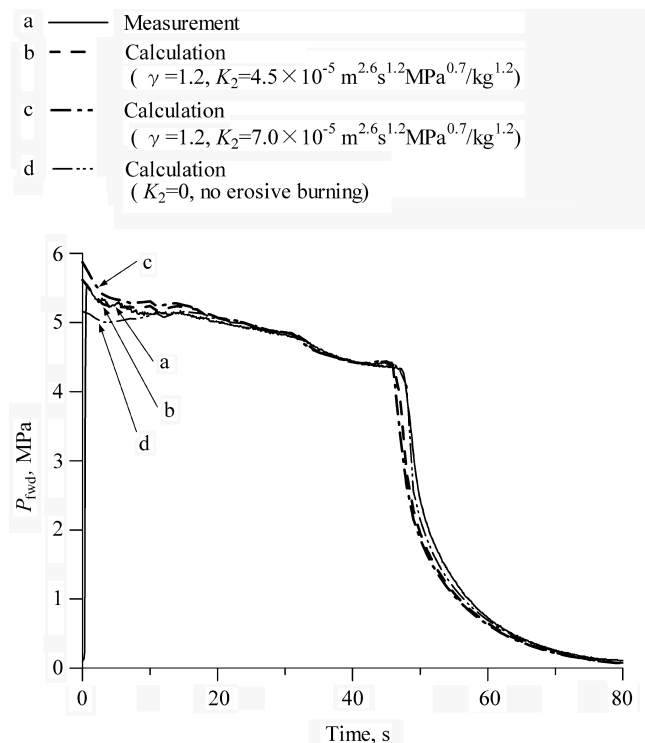


Fig. 17 Measured and calculated pressure histories of the M-14 motor.

DSM. Mass flux in the port, pressure, port hydraulic diameter, and dimensionless threshold were incorporated in the equation.

Parameters in the correlative equation were first determined based on the DSM test results, and finally improved by taking into account the experimental results of both cylindrical test motors and practical full-scale motors.

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